Hall Ticket Number:

MR-22

Code: B425CM5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2025 / January 2026

Object Oriented Programming through Java

(Computer Science and Engineering (AI and ML))

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What is inheritance in Java?	1	1	1	1
1. b)	What is garbage collection in Java?	1	2	1	3
1. c)	Define a package in Java.	1	1	2	2
1. d)	What is the CLASSPATH variable used for?	1	2	2	2
1. e)	What is autoboxing in Java?	1	1	3	3
1. f)	What is the use of 'finally' block in exception handling?	1	2	3	3
1. g)	List types of events in Java.	1	1	4	1
1. h)	State the purpose of the 'Delegation Event Model'.	1	2	4	2
1. i)	List two limitations of AWT.	1	1	5	2
1. j)	Define the life cycle of an applet.	1	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Write a Java program to demonstrate exception handling using try-catch blocks.	5	1	1	2
2. b)	Discuss the concept of recursion in Java with a suitable example program.	5	2	1	3
	OR				
2. c)	Explain how inheritance works in Java with a program example.	5	1	1	2
2. d)	Write a Java program to create and manipulate an array of objects.	5	1	1	4
3. a)	Write a Java program to create and use an interface.	5	1	2	3
3. b)	Explain the costs and benefits of inheritance with examples.	5	3	2	2
	OR				
3. c)	Write a Java program to demonstrate the use of the 'final' keyword with inheritance.	5	1	2	4
3. d)	Describe how abstract classes differ from interfaces with an example.	5	2	2	3
4. a)	Explain the life cycle of a thread with examples.	5	3	3	2
4. b)	Write a Java program to demonstrate the use of generics with collections.	5	4	3	4
	OR				
4. c)	Describe the exception hierarchy in Java with examples.	5	2	3	3
4. d)	Develop a Java program to read and write strings using java.io.	5	3	3	4
5. a)	Discuss the different types of layout managers in Java with examples.	5	2	4	4
5. b)	Write a Java program to create a user interface with labels, buttons, and text components.	5	3	4	4
	OR				
5. c)	Explain the use of the 'CardLayout' manager with an example program.	5	3	4	2
5. d)	Explain the MVC architecture in Swing and how it addresses the limitations of AWT.	5	2	4	2

P.T.O.

6. a)	Discuss the architecture and components of Swing with examples.	5	4	5	4
6. b)	Write a Java program to create a user interface with JLabels, JTextFields, and JButtons.	5	3	5	4
	OR				
6. c)	Write a Java program to create a tabbed pane and add tabs with different components in each tab.	5	4	5	4
6. d)	Explain the life cycle of an applet with an example program that demonstrates each phase.	5	2	5	4

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome

Hall Ticket Number:

MR-22

Code: B425IT5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2025 / January 2026

Java Programming

(Information Technology)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Name two key features of Java.	1	1	1	1
1. b)	Give an example of a primitive data type in Java.	1	1	1	1
1. c)	Differentiate Built-in and custom exception in Java?	1	2	2	1
1. d)	Name the classes used for file handling in Java.	1	1	2	1
1. e)	Name the utility classes in Java.	1	1	2	1
1. f)	Explain the purpose of the Comparable interface in Java.	1	2	2	1
1. g)	Provide an example of a process in a computer system.	1	2	3	1
1. h)	Explain the purpose of JDBC drivers in Java Database Connectivity.	1	2	3	1
1. i)	Explain the limitations of AWT in comparison to Swing.	1	2	4	1
1. j)	Discuss the advantages of using the Delegation event model over other event handling models.	1	2	4	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Discuss at least five key features of Java and how they contribute to its popularity.	5	2	1	1
2. b)	Describe the significance of access modifiers in Java and provide examples of their usage.	5	2	1	2
	OR				
2. c)	Explain the role of variables in Java and illustrate with examples.	5	2	1	1
2. d)	Discuss different types of operators and their usage in Java with examples.	5	2	1	1
3. a)	Describe the RandomAccessFile class in Java and its significance in file handling operations. Provide examples of its usage and functionalities.	5	2	2	3
3. b)	Explain the purpose and usage of byte streams in Java. Discuss various types of byte streams and their functionalities.	5	2	2	2
	OR				
3. c)	Explain the purpose and usage of exception handlers in Java. Discuss the try-catch-finally block with examples.	10	2	2	1
4. a)	Explain the process of importing packages in Java. Discuss the benefits of using import statements. Provide examples.	5	2	2	3
4. b)	Describe the Formatter class in Java and its role in formatting text. Discuss its methods and functionalities. Provide examples.	5	2	2	2
	OR				
4. c)	Describe the concept of collections in Java and their role in storing and manipulating groups of objects.	5	2	2	1,2
4. d)	Discuss how sorting is implemented using the Comparable and Comparator interfaces. Provide examples for both approaches.	5	2	2	2,3

P.T.O.

5. a)	Analyze the Java thread life cycle and discuss how understanding it helps in writing efficient multithreaded programs.	5	4	3	2,3
5. b)	Design a JDBC application that demonstrates the basic steps involved in connecting to a database, creating a new database, and creating a table.	5	3	3	2,3
	OR				
5. c)	Dcvclop a multithrcadcd Java program that demonstrates the use of synchronized methods or blocks to ensure thread safety.	5	3	3	3
5. d)	Compare and contrast the different types of JDBC drivers. Discuss their advantages, disadvantages, and suitable use cases.	5	4	3	1
6. a)	Design a Swing application following the MVC architecture. Discuss the role of each component (Model, View, Controller).	5	3	4	1
6. b)	Compare and contrast the Delegation event model with other event handling models in Java. Discuss the advantages and disadvantages of each model.	5	4	4	1,2
	OR				
6. c)	Explore factors such as flexibility, ease of use, and responsiveness in choosing the appropriate layout manager for different UI requirements.	5	4	4	2,3
6. d)	Design a Java application that demonstrates event handling for both mouse and keyboard events.	5	3	4	1,3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome